

AI in Finance: Stock Market Prediction Using AI and ML

A Systematic Literature Review

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Abstract

This paper presents a systematic review of the literature on the application of machine learning and artificial intelligence to predict stock market movements. The financial market process is highly volatile and non-linear. Therefore, several studies have attempted to use advanced data science techniques such as Random Forest, Support Vector Machine, Artificial Neural Network, Long Short-Term Memory, and hybrid models to predict stock market trends with better accuracy. The results indicate that while the use of artificial intelligence and machine learning leads to better predictions, higher accuracy is still challenging mainly because of the volatility of the market. The study recommends hybrid and ensemble models to better analyze multiple variables influencing stock market trends. This review highlights research opportunities and directions for future studies on the topic.

1. Introduction

Right from the start, investor mood shapes how markets move, showing what people think about future profits [1]. Instead of steady patterns, price shifts often jump around without clear cause, reacting fast to news or rumors [2]. Money flows into stocks because companies need funds to grow, which helps entire economies expand over time. Still, small changes in data or policy can ripple through trading activity in ways that surprise even experts. Over the years, scholars have looked closely at trends using mathematical tools, trying to spot hidden order beneath chaos. Globally, institutions watch these swings carefully since outcomes affect jobs, spending, and savings alike. Even with powerful computers, predictions fail regularly due to too many moving parts influencing results. Because emotions mix with facts during trades, logic alone rarely explains sudden market turns [3].

Back then, picking stocks meant digging into numbers or studying charts. Looking at balance sheets, profits, and income streams – this way tries to find what a business is truly worth. Charts tell another story, one built on past prices and how much was traded, searching for familiar rhythms [2]. Even though these methods work sometimes, today's markets spit out too much information too fast for them to keep up. When emotions swing wildly or something unexpected hits, those old tools often fall short. Their blind spots grow bigger when change arrives without warning [4].

Twenty years of fast progress in artificial intelligence together with machine learning have reshaped how forecasts are made in the stock market. Traditional number-crunching approaches fall short where these systems thrive – spotting hidden links inside massive piles of information [3]. They learn from what happened before, adjusting when markets shift direction. Instead of relying on one method, researchers mix different tools like neural nets, support vector machines, random forest setups, LSTM cells, or combined strategies. Each handles specific kinds of math challenges tied to pricing trends and risk signals. Deeper layers of processing now pull meaning from messy sources such as tweets or headlines through emotion tracking tricks baked into advanced networks [1]. These upgrades let predictions run deeper than older formulas ever allowed.

Despite the proliferation of research in this field, there is no consensus on a single “optimal” method for stock market prediction. Performance varies significantly depending on the choice of algorithm, data

preprocessing, feature selection, and evaluation metrics [5]. Moreover, while numerous studies introduce specific models, comprehensive reviews that synthesize findings across different methodologies and identify future research directions remain relatively scarce [6, 7].

This paper provides a systematic review of recent research on AI and machine learning in stock market forecasting. It categorizes the various methodologies employed, examines the types of data used, and analyzes the performance metrics reported in the literature. By identifying the strengths and weaknesses of current approaches, this work aims to highlight critical research gaps and suggest directions for future study [3, 6].

The remainder of this paper is organized as follows. Section 2 presents the background and foundational concepts relevant to the topic. Section 3 reviews the machine learning methodologies identified in the literature. Section 4 provides a comparative analysis of the selected studies. Section 5 identifies key research gaps. Section 6 suggests directions for future work. Section 7 concludes the paper.

2. Background

The stock market is a dynamic and essential component of the global financial system, reflecting economic health and providing opportunities for wealth accumulation [8]. However, predicting price movements remains a formidable challenge due to the high degree of uncertainty and the influence of diverse factors, including corporate earnings, geopolitical events, and investor psychology [1, 2]. The complexity of these interactions makes the market inherently difficult to forecast [4].

Traditional forecasting methods are generally categorized into fundamental and technical analysis. Fundamental analysis focuses on economic and financial indicators to determine a stock's intrinsic value [9]. Technical analysis, on the other hand, utilizes historical price patterns and indicators such as the Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), and Simple Moving Averages (SMA) to predict future price movements [2]. While fundamental analysis seeks to determine what a stock should be worth, technical analysis attempts to identify where the price is likely to go based on past trends.

Two major theories have historically guided the understanding of market predictability: the Efficient Market Hypothesis (EMH) and the Adaptive Market Hypothesis (AMH). EMH suggests that stock prices fully reflect all available information, making it impossible to consistently outperform the market [10]. In contrast, AMH posits that market efficiency is not static but evolves over time, allowing for periods of predictability when certain conditions are met [9].

The availability of high-performance computing and massive financial datasets has shifted the focus toward AI-driven forecasting [2]. These systems can identify patterns in historical data without the need for explicit programming, making them well-suited for the complex and non-linear nature of financial markets [11]. Machine learning tasks in this domain typically involve either classification (e.g., predicting “buy,” “sell,” or “hold” signals) or regression (e.g., predicting specific price values) [12].

Prominent techniques in machine learning research include Artificial Neural Networks (ANN), Support Vector Machines (SVM), and Random Forests. Additionally, Long Short-Term Memory (LSTM) networks – a specialized type of Recurrent Neural Network (RNN) – have become popular for their ability to capture long-term dependencies in time-series data [13, 14]. Hybrid and ensemble methods, which combine the outputs of multiple models, are also increasingly used to improve forecast reliability and reduce error [11].

Recent research has also begun to incorporate sentiment analysis, using NLP to extract market mood from social media and financial news. This multi-source approach represents a significant shift in stock market forecasting, as it acknowledges the impact of public opinion on trading behavior [1, 15].

3. Literature Review

Over the past two decades, there has been a significant increase in the application of artificial intelligence for stock market forecasting. This section reviews selected studies categorized by their primary methodologies: classical machine learning, deep learning, hybrid/ensemble methods, and multi-source sentiment analysis.

3.1 Classical Machine Learning Methods

Early research in AI-driven forecasting established the baseline for modern techniques by testing fundamental machine learning models. Mokhtari et al. (2021) conducted a broad assessment using technical indicators and financial fundamentals to predict Apple Inc. (AAPL) stock prices. Their study compared several models, including Logistic Regression, K-Nearest Neighbors (KNN), Random Forest, and SVM. Notably, Linear Regression achieved high accuracy for price prediction, with R^2 values nearing ideal levels, outperforming LSTM models in that specific context due to lower error metrics (MAPE of 1.56 vs. 2.99). However, for sentiment classification, SVM proved most effective, reaching an accuracy of 75.5% and an AUC of 0.76. Despite these results, the authors noted that AI models still face challenges in maintaining consistent performance over extended periods [9].

Akhtar et al. (2022) examined the performance of Random Forest and SVM on historical price data, reporting an aggregate accuracy of 80.3%. Their findings emphasized that Random Forest slightly outperformed SVM (80.8% vs. 78.7%) and highlighted the critical importance of data preprocessing. Without rigorous cleaning and normalization, model stability and forecast accuracy were significantly diminished [12].

Sheth and Shah (2023) provided a detailed comparison of ANN, SVM, and LSTM. Their results showed that ANNs consistently achieved approximately 90% accuracy due to their ability to detect complex non-linear trends. SVM performance varied between 60% and 91%, depending on the kernel selection and parameter tuning. Interestingly, integrating SVM with a Genetic Algorithm for feature selection improved precision to 93.7%, demonstrating the value of optimization techniques [13].

3.2 Deep Learning Models

As deep learning evolved, LSTM networks became a preferred choice for financial time-series forecasting. LSTMs address the vanishing gradient problem inherent in traditional RNNs, allowing them to capture long-term dependencies in sequential data [11].

Sonkavde et al. (2023) analyzed various machine learning and deep learning models using Indian stock datasets (TAINIWALCHM and AGROPHOS). While LSTM showed promise, XGBoost achieved superior results in their sample, with an R^2 of 0.984 and a lower RMSE. The study concluded that while LSTMs are powerful for temporal patterns, their performance is highly dependent on large volumes of data and precise hyperparameter tuning [11].

Sheth and Shah (2023) further discussed the architectural advantages of LSTMs, noting that their gated cell structure – comprising input, forget, and output gates – allows for selective information retention. Their review reported LSTM accuracies ranging from 57% to over 90%, with Attention-based LSTM variants often producing the most reliable trading signals [13].

3.3 Ensemble and Hybrid Methods

To overcome the limitations of individual algorithms, researchers have increasingly developed ensemble and hybrid models. Sonkavde et al. (2023) proposed an ensemble framework combining Random Forest, XGBoost, and LSTM. This hybrid model outperformed all individual algorithms, achieving an R^2 of 0.992. The authors attributed this success to the model's ability to simultaneously handle high-dimensional data, sequential learning, and temporal dependencies [11].

Venkatarathnam et al. (2024) explored hybrid modeling using Genetic Algorithms (GA) to optimize SVM and ANN parameters for NSE-listed stocks. Their results confirmed that GA-optimized models consistently outperformed standalone versions. Additionally, they applied the ARIMA model for short-term forecasting, finding that while it was effective for immediate trends, its accuracy declined significantly over longer time horizons [16].

3.4 Sentiment Analysis and Multi-Source Approaches

A growing body of research incorporates textual and sentiment-based information, recognizing the influence of news and public opinion on market movements. Mokhtari et al. (2021) demonstrated that Twitter sentiment could serve as a proxy for fundamental analysis, though they noted the difficulty of converting unstructured text into precise trading signals [9].

Sonkavde et al. (2023) highlighted that combining sentiment data with historical prices consistently improves model accuracy. Training LSTMs on both price values and sentiment scores (encoded via Word2Vec or N-Grams) led to substantial performance gains. They also emphasized the necessity of filtering "fake news" and spam to maintain data quality [11].

3.5 Meta-Review Findings

Lin and Marques (2024) synthesized findings from ten major literature reviews, covering over 379 individual studies. Their meta-analysis confirmed that SVM, LSTM, and ANN remain the most widely used AI methods in the field. Historical closing prices are the dominant input source (used in 66% of studies), followed by technical indicators and sentiment data. A key gap identified was the lack of studies that simultaneously integrate social media sentiment, financial news, historical prices, and macroeconomic data, suggesting that such multi-source hybrid approaches represent the next frontier in research [14].

4. Comparative Analysis

Table 1 summarizes the technique, dataset, reported accuracy, contribution, and limitation of sixteen selected studies reviewed in this paper.

Table 1: Comparative analysis of selected studies on AI/ML-based stock market prediction.

Paper	Technique	Dataset	Accuracy	Contribution	Limitation
Sizan et al. (2023) [6]	Random Forest, Gradient Boosting, Logistic Regression	Historical stock prices (Yahoo Finance, Google Finance), financial indicators, news data	Not specified	Provides a structured comparison of ML models for U.S. stock forecasting and highlights practical implications for investors	Limited exploration of deep learning models; results depend heavily on selected features
Sonkavde et al. (2023) [11]	ML & DL (Random Forest, XGBoost, LSTM, ensemble models)	Real stock datasets (TAINI-WALCHM, AGROPHOS) + financial features	Ensemble (RF + XGBoost + LSTM) outperformed individual models	Proposes hybrid ensemble framework and systematic review of ML/DL models	High computational complexity; interpretability is low
Akhtar et al. (2022) [12]	Random Forest, Support Vector Machine (SVM)	Historical stock price dataset (pre-processed)	~80.3% prediction accuracy	Emphasizes importance of preprocessing and model tuning for better performance	Limited generalization due to dataset size and lack of external factors
Rouf et al. (2021) [3]	ML, text analytics, ensemble methods	Literature datasets (2011–2021 studies)	Not applicable (survey-based)	Provides decade-long overview and identifies emerging trends like sentiment analysis	No experimental validation; lacks real-world performance metrics
Kumar & Sarangi (2022) [4]	ANN, SVM, Neural Networks	Review of 30 research studies	Not specified	Identifies ANN and NN as most widely used techniques	No quantitative comparison; lacks experimental benchmarking
Bansal et al. (2022) [17]	KNN, Linear Regression, SVR, Decision Tree, LSTM	Indian stock market data (12 companies, 7 years)	LSTM produced highest accuracy among models	Provides empirical comparison between ML and DL methods on real-world data	Dataset limited geographically; may not generalize globally
Khan et al. (2022) [1]	ML + Deep Learning + Sentiment Analysis	Social media (Twitter), financial news, stock data	Up to 83.22% (Random Forest ensemble)	Demonstrates importance of sentiment data in improving prediction accuracy	Data noise and bias from social media; requires heavy preprocessing
Sheth & Shah (2023) [13]	ANN, SVM, LSTM	General stock datasets	ANN showed best results for nonlinear patterns	Explains strengths of neural networks in handling complex relationships	LSTM requires large datasets; computational cost is high
Lin & Marques (2024) [14]	SVM, LSTM, ANN (systematic review of reviews)	379+ studies	Not applicable	Identifies dominant AI techniques and highlights research gaps	No practical experimentation
Strader et al. (2020) [8]	ANN, SVM, Genetic Algorithms, Hybrid models	Literature-based	Not applicable	Categorizes ML techniques into major research groups and proposes future directions	Focuses on classification rather than performance comparison
Chopra & Sharma (2021) [10]	Neural networks, hybrid AI techniques	148 research papers	Not applicable	Provides taxonomy of AI techniques and research agenda	No empirical validation; theoretical focus
Subasi et al. (2021) [2]	Multiple ML classifiers	NASDAQ, NYSE, FTSE, Nikkei datasets	Varies across models	Compares ML models across global markets	Performance sensitive to dataset variations
Kompella & Chilukuri (2020) [18]	Random Forest + Sentiment Analysis	Historical stock data + news sentiment	Not specified	Shows effectiveness of Random Forest combined with sentiment data	Cannot achieve consistent accuracy due to volatility

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Paper	Technique	Dataset	Accuracy	Contribution	Limitation
Ampomah et al. (2021) [19]	Gaussian Naïve Bayes + PCA/LDA	Stock price datasets	GNB + LDA performed best	Explores feature scaling and extraction impact on ML performance	Limited comparison with advanced ML/DL models
Koukaras et al. (2022) [15]	ML + Sentiment Analysis (SVM, KNN, NB, etc.)	Twitter, Stock-Twits, Yahoo Finance	F-score: 76.3%, AUC: 67%	Integrates sentiment analysis with ML for better predictions	Dependent on sentiment data quality
Venkatarathnam et al. (2024) [16]	AI & ML techniques	General financial datasets	Not specified	Highlights real-world impact of AI in stock trading	Lacks experimental results

5. Research Gaps

Despite significant advancements in AI-driven stock market prediction, several critical research gaps remain. Addressing these limitations is essential for improving model robustness, generalizability, and practical utility in real-world trading environments [3, 10].

1. Limited Multi-Source Data Integration

A significant portion of current research relies heavily on historical price data and a narrow set of technical indicators. While these features are readily available, they represent only a fraction of the factors influencing market dynamics. Fundamental data – such as earnings reports, balance sheets, and macroeconomic indicators – is frequently neglected due to its unstructured and inconsistent nature. Furthermore, although sentiment analysis of social media and news has gained traction, the simultaneous integration of technical, fundamental, and sentiment data within a single predictive framework remains rare. Developing models that can effectively synthesize these diverse data types is a key area for future exploration [11, 14].

2. Overfitting and Lack of Generalizability

A recurring challenge in the literature is the tendency of complex models, such as LSTMs and deep ensembles, to overfit historical patterns, leading to poor performance when market conditions shift [13, 17]. Many studies validate their models on limited datasets or specific time periods without testing for cross-market or cross-temporal generalizability. Consequently, the predictive power of these models in unseen market regimes remains largely unproven [2].

3. Absence of Standardized Evaluation Metrics

The lack of a unified evaluation framework complicates the comparison of results across different studies. Researchers employ a wide array of metrics, including accuracy, F1-score, RMSE, MAE, and R^2 , often focusing on different assets or indices. This inconsistency makes it difficult to establish benchmarks and verify the relative superiority of emerging techniques [8, 14].

4. Short-Term vs. Long-Term Forecasting

Most existing research focuses on short-term price movements, such as daily or weekly forecasts. While valuable for high-frequency trading, these models are less applicable to long-term investment planning.

Developing reliable AI methodologies for extended time horizons remains a significant challenge due to the increasing noise and uncertainty associated with long-term market trends [11, 17].

5. Under-Researched Emerging Markets

The majority of literature focuses on developed markets, particularly in the U.S., China, and Western Europe. Emerging economies in South Asia, Africa, and Latin America are often overlooked, despite their growing influence in the global financial system. These markets present unique challenges, such as lower liquidity and less consistent data reporting, which require tailored AI approaches [8, 16].

6. Incomplete Exploration of Hybrid and Ensemble Methods

While hybrid models often outperform standalone algorithms, many potential combinations remain unexplored. For instance, integrating statistical models like ARIMA with deep learning architectures like LSTM, or utilizing Genetic Algorithms for hyperparameter optimization in complex neural networks, offers promising avenues for improving forecast accuracy that have yet to be fully realized [3, 11].

7. Neglect of Transaction Costs and Market Constraints

Many models report high theoretical accuracy but fail to account for practical trading constraints, such as transaction fees, slippage, and liquidity. In a real-world setting, these factors can significantly erode predicted profits. Future research must incorporate realistic trading simulations to validate the economic viability of AI-driven strategies [5].

8. Model Interpretability and the “Black Box” Problem

Advanced deep learning models often function as “black boxes,” providing little insight into the underlying reasoning for their predictions. In the financial sector, where risk management and regulatory compliance are paramount, the lack of interpretability is a major barrier to adoption. Developing “Explainable AI” (XAI) techniques tailored for financial forecasting is crucial for building trust among investors and regulators [5, 10].

6. Future Directions

Future research should focus on enhancing the practical applicability and robustness of AI models in financial markets. Key directions include:

- **Integration of Alternative Data Sources:** Beyond traditional price and volume data, incorporating alternative sources such as satellite imagery, web search trends, and corporate filings could provide a more comprehensive view of market drivers.
- **Long-Term Forecasting Models:** Shifting focus toward multi-month and multi-year horizons will be beneficial for institutional investors and long-term financial planning.
- **Cross-Market Analysis:** Expanding research to include emerging markets will help validate the universality of AI models and uncover region-specific market dynamics.
- **Explainable AI (XAI):** Developing models that offer transparent reasoning for their predictions is essential for regulatory compliance and investor confidence.

- **Realistic Backtesting:** Incorporating transaction costs, market impact, and liquidity constraints into model evaluations will bridge the gap between theoretical research and actual trading performance.

The continuous evolution of AI and machine learning presents significant opportunities for the financial sector. By addressing the identified research gaps and focusing on these future directions, researchers can develop more reliable and transparent tools for stock market prediction [11, 14].

7. Conclusion

This systematic review has explored the application of artificial intelligence and machine learning in stock market prediction, synthesizing insights from numerous academic studies. The literature reveals a clear transition from traditional statistical methods to sophisticated deep learning and ensemble architectures. While the inherent volatility and complexity of financial markets remain significant challenges, the integration of advanced algorithms has demonstrated superior performance in identifying non-linear patterns within large datasets [5, 17].

Key findings indicate that ensemble models – particularly those combining Random Forest, XGBoost, and LSTM – consistently achieve higher accuracy by leveraging the complementary strengths of different techniques [6, 11]. Furthermore, the inclusion of sentiment analysis from news and social media has been shown to enhance predictive power, suggesting that market sentiment is a critical factor alongside historical price data [1, 15].

Despite these advancements, several hurdles persist. Many models suffer from overfitting and lack generalizability across different market regimes. The absence of standardized evaluation metrics and the “black box” nature of deep learning architectures further complicate the practical adoption of AI in finance. Future research must address these issues by focusing on multi-source data integration, long-term forecasting, and the development of explainable AI frameworks.

In conclusion, while no single method provides a definitive solution for stock market forecasting, the continued development of AI and machine learning offers promising pathways for improving market analysis and investment strategies. By addressing current research gaps and emphasizing model transparency and robustness, the field can move toward more reliable and actionable financial predictions [8, 10, 14].

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